

significant potential especially as a non-lethal weapon. PBWs at this time are apt to remain in the science fiction domain, as the weight and cost as yet do not justify achievable military effect.

Though technology needed for PBWs exists and seems feasible, it is too impractical for miniaturisation and operational deployment. There are no known operational PBWs and, as of now, such weapons exist only in science fiction and artists' imagination (Figs 3 and 4). Fig. 5 shows yet another PBW called Disruptor known in science fiction.

PBW's

A PBW is a form of DEW that uses atomic or subatomic particles accelerated to the speed of light or near speed of light with the help of powerful electric and magnetic fields in a particle accelerator. The particles are directed to deliver a fraction of their kinetic energy to the intended target, thereby causing severe damage due to disruption of its atomic structure. It is characterised by beam energy in electron-volts, beam current in amperes and beam power in watts.

PBW's come in two primary types: charged-particle weapons and neutral-particle weapons. When it comes to military application of these different types of PBWs, charged-particle weapons are endo-atmospheric, while neutral-particle weapons are exo-atmospheric.

At the core of a PBW is the particle accelerator. It is also the most complex part of the beam weapon and is built using a linear electric field to accelerate charged particles similar to Gauss or coil gun or an induction linear accelerator system.

Induction linear accelerator consists of a simple non-resonant structure where drive voltage is applied to an axially symmetric gap that encloses a toroidal ferromagnetic material. Change in

flux in the magnetic core induces an axial electric field that provides particle acceleration.

Characteristic parameters

A particle beam consists of protons, electrons or neutral atoms flowing with real or imaginary current. It is characterised by beam energy, current and power. Beam energy is expressed in mega electron-volts (MeV). One eV is the kinetic energy of an electron that has been accelerated by an electric potential of one volt. Particle beam energy is characterised by the energy of a typical particle of the beam as all particles in a beam will have been



Fig. 5: PBW called Disruptor in science fiction

accelerated to the same velocity. A PBW capable of inflicting serious damage to a target 1000km away in space would typically require beam energy of 1GeV.

An estimate of the number of charged particles in the beam can be made from the magnitude of beam current. It is possible to assign a current to the particle beam, assuming that each particle has a charge quantum equal to that of an electron even if the charged particle was the neutral atom. Beam current for the possible beam weapon described above would typically be 1000 amperes.

Power of a particle beam is the rate at which beam energy is transported, which is also indicative of the rate at which it can deposit energy into a target. As an analogy to electric circuits, the particle beam in watts is equal to the product of

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